

“PAPER_{0:D3}” -f [int]# PAPER #86 □ Ug4 AGN Feedback: 8-Parameter UQFF Formula

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

Source Data: test_Ug4_validation.py, Ug4StarBlackHoleCalculator, UQFFConstants-Database, SAGITTARIUS_A_STAR_2025

Index Slot: §1.11 Black Hole Physics & Hawking Radiation,
\$n = [int]# PAPER #86 □ Ug4 AGN Feedback: 8-Parameter UQFF Formula

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

Source Data: test_Ug4_validation.py, Ug4StarBlackHoleCalculator, UQFFConstants-Database, SAGITTARIUS_A_STAR_2025

Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_086

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun□Sgr A* system at 27,000 ly, the validator test_Ug4_validation.py computes $Ug4 = 3.352941 \times 10^{22} \text{ J/m}^3$ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants (? = $5.0 \times 10^4 \text{ day}^{-1}$, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole□stellar system boundary □ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{decay}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_bh	8.55×10^{36} kg	EHT 2024-25
Orbital distance	d_g	2.55×10^{20} m	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_AGN	1.0 (quiescent)	Amplification factor
Decay constant	a	?/t_orb	Tied to ? = 0.0005/day
SCm density	[SCm]	0.99	Superconductive mode
UA density	[UA]	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun-Sgr A*** system at t=0, t_n=0, A_AGN=1.0:

$$Ug_4^{\text{Sun-SgrA}^*}(t = 0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: e^{-at}

The Ug4 baseline decays exponentially with the UQFF ? parameter:

$$f_{decay}(t) = e^{-\kappa t}$$

With ? = 0.0005/day = $5.787 \times 10^{??} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay e^{-at}) □ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) □ PASS (Ug4 increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: cos(pt_n)

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle cos(pt_n)) □ PASS

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_{bh} (kg)	d_{g} (m)	$\text{Ug4}(t=0)$ (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) □ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full $F_{U_{Bi_i}}$ master equation
- Real-time Ug4 evaluation: triggered by change in M_{bh} , d_g , or t_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	$Ug4 = 3.352941 \times 10^{22}$ at ($t=0$, $t_n=0$)	PASS
2. Temporal decay	e^{-at} ? monotonic decrease	PASS
3. AGN feedback	$A_{AGN} \times f_{SCm}$ amplification	PASS
4. Negative time	$Ug4 > \text{baseline}$ (pre-collapse regime)	PASS
5. Temporal cycle	$\cos(pt_n)$ recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full $F_{U_{Bi_i}}$	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: `test_Ug4_validation.py` | `Ug4StarBlackHoleCalculator` | `SAGITTARIUS_A_STAR_2025`
| 7 tests PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun–Sgr A* system at 27,000 ly, the validator `test_Ug4_validation.py` computes $Ug4 = 3.352941 \times 10^{22} \text{ J/m}^3$ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^4 \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole–stellar system boundary – analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From `Ug4StarBlackHoleCalculator` and `UQFFConstantsDatabase`:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{decay}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_bh	8.55×10^{36} kg	EHT 2024-25
Orbital distance	d_g	2.55×10^{20} m	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_AGN	1.0 (quiescent)	Amplification factor
Decay constant	a	?/t_orb	Tied to ? = 0.0005/day
SCm density	[SCm]	0.99	Superconductive mode
UA density	[UA]	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun-Sgr A*** system at t=0, t_n=0, A_AGN=1.0:

$$Ug_4^{\text{Sun-SgrA}^*}(t = 0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: e^{-at}

The Ug4 baseline decays exponentially with the UQFF ? parameter:

$$f_{decay}(t) = e^{-\kappa t}$$

With ? = 0.0005/day = $5.787 \times 10^{??} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay e^{-at}) □ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) □ PASS (Ug4 increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: cos(pt_n)

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle cos(pt_n)) □ PASS

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_{bh} (kg)	d_{g} (m)	$\text{Ug4}(t=0)$ (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) □ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full $F_{U_{Bi_i}}$ master equation
- Real-time Ug4 evaluation: triggered by change in M_{bh} , d_g , or t_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	$Ug4 = 3.352941 \times 10^{22}$ at ($t=0$, $t_n=0$)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	$A_{AGN} \times f_{SCm}$ amplification	PASS
4. Negative time	$Ug4 >$ baseline (pre-collapse regime)	PASS
5. Temporal cycle	$\cos(pt_n)$ recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full $F_{U_{Bi_i}}$	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: *test_Ug4_validation.py* | *Ug4StarBlackHoleCalculator* | *SAGITTARIUS_A_STAR_2025*
| 7 tests PASS .Groups[1].Value □ Ug4 AGN Feedback: 8-Parameter UQFF Formula

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($? = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: *test_Ug4_validation.py*, *Ug4StarBlackHoleCalculator*, *UQFFConstants-Database*, *SAGITTARIUS_A_STAR_2025*

Index Slot: §1.11 Black Hole Physics & Hawking Radiation,

$\$n = [\text{int}] \# \text{"PAPER_}\{0:D3\}" -f [\text{int}] \# \text{PAPER \#86} \square Ug4 \text{ AGN Feedback: 8-Parameter UQFF Formula}$

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($? = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: *test_Ug4_validation.py*, *Ug4StarBlackHoleCalculator*, *UQFFConstants-Database*, *SAGITTARIUS_A_STAR_2025*

Index Slot: §1.11 Black Hole Physics & Hawking Radiation,

$\$n = [\text{int}] \# \text{PAPER \#86} \square Ug4 \text{ AGN Feedback: 8-Parameter UQFF Formula}$

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($? = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: *test_Ug4_validation.py*, *Ug4StarBlackHoleCalculator*, *UQFFConstants-Database*, *SAGITTARIUS_A_STAR_2025*

Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_086

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun–Sgr A* system at 27,000 ly, the validator test `Ug4_validation.py` computes $Ug_4 = 3.352941 \times 10^{22} \text{ J/m}^3$ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, *M87*, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^{\tau^1}$, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole–stellar system boundary – analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From `Ug4StarBlackHoleCalculator` and `UQFFConstantsDatabase`:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{decay}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_{bh}	$8.55 \times 10^36 \text{ kg}$	EHT 2024-25
Orbital distance	d_g	$2.55 \times 10^{20} \text{ m}$	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_{AGN}	1.0 (quiescent)	Amplification factor
Decay constant	a	τ/t_{orb}	Tied to $\tau = 0.0005/\text{day}$
SCm density	$[SCm]$	0.99	Superconductive mode
UA density	$[UA]$	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun–Sgr A*** system at $t=0$, $t_n=0$, $A_{AGN}=1.0$:

$$Ug_4^{\text{Sun-SgrA}^*}(t = 0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: $e^{-\kappa t}$

The Ug4 baseline decays exponentially with the UQFF κ parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-5} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay $e^{-\kappa t}$) $\square$ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) $\square$ **PASS** (Ug4 increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: $\cos(\pi t_n)$

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle $\cos(pt_n)$) □ PASS

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_{bh} (kg)	d_g (m)	$Ug4(t=0)$ (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) □ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full $F_{U_{Bi_i}}$ master equation
 - Real-time Ug4 evaluation: triggered by change in M_{bh} , d_g , or t_n
 - Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures
-

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	$Ug4 = 3.352941 \times 10^{22}$ at ($t=0$, $t_n=0$)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	$A_{\text{AGN}} \times f_{\text{SCm}}$ amplification	PASS
4. Negative time	$Ug4 >$ baseline (pre-collapse regime)	PASS
5. Temporal cycle	$\cos(pt_n)$ recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full $F_{U_{Bi_i}}$	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: `test_Ug4_validation.py` | Ug4StarBlackHoleCalculator | SAGITTARIUS_A_STAR_2025
 | 7 tests PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun-Sgr A* system at 27,000 ly, the validator `test_Ug4_validation.py` computes $Ug4 = 3.352941 \times 10^{22}$ J/m³ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay

(e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, *M87*, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^{-1}$, $[\text{SCm}] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole–stellar system boundary – analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$Ug_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SCm}]) = \frac{G^2 M_{\text{bh}}^2}{c^4 d_g^6} \cdot \frac{[\text{SCm}]}{1 + [\text{UA}]} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_{bh}	$8.55 \times 10^36 \text{ kg}$	EHT 2024-25
Orbital distance	d_g	$2.55 \times 10^{20} \text{ m}$	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_{AGN}	1.0 (quiescent)	Amplification factor
Decay constant	a	τ/t_{orb}	Tied to $\tau = 0.0005/\text{day}$
SCm density	$[\text{SCm}]$	0.99	Superconductive mode
UA density	$[\text{UA}]$	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun–Sgr A*** system at $t=0$, $t_n=0$, $A_{\text{AGN}}=1.0$:

$$Ug_4^{\text{Sun-SgrA}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: $e^{-\kappa t}$

The Ug4 baseline decays exponentially with the UQFF κ parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-10} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay $e^{-\kappa t}$) $\square$ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) $\square$ **PASS** (Ug4 increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: $\cos(\pi t_n)$

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle $\cos(\pi t_n)$) $\square$ **PASS**

7. Three Pre-Defined Systems

From test_Ug4_validation.py:

System	M_bh (kg)	d_g (m)	Ug4(t=0) (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) □ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full F_U_Bi_i master equation
- Real-time Ug4 evaluation: triggered by change in M_bh, d_g, or t_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	Ug4 = 3.352941×10^{22} at (t=0, t_n=0)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	A_AGN $\times$ f_SCm amplification	PASS
4. Negative time	Ug4 > baseline (pre-collapse regime)	PASS
5. Temporal cycle	cos(pt_n) recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full F_U_Bi_i	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: test_Ug4_validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUS_A_STAR_2025
| 7 tests PASS .Groups[1].Value □ Ug4 AGN Feedback: 8-Parameter UQFF Formula

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

Source Data: test_Ug4_validation.py, Ug4StarBlackHoleCalculator, UQFFConstants-Database, SAGITTARIUS_A_STAR_2025

Index Slot: §1.11 Black Hole Physics & Hawking Radiation, "PAPER_{0:D3}" -f [int]# PAPER #86 □ Ug4 AGN Feedback: 8-Parameter UQFF Formula

Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

Source Data: test_Ug4_validation.py, Ug4StarBlackHoleCalculator, UQFFConstants-Database, SAGITTARIUS_A_STAR_2025
Index Slot: §1.11 Black Hole Physics & Hawking Radiation,
 \$n = [int]\$ PAPER #86 □ Ug4 AGN Feedback: 8-Parameter UQFF Formula
Title: Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole□Host Galaxy Coupling
Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Source Data: test_Ug4_validation.py, Ug4StarBlackHoleCalculator, UQFFConstants-Database, SAGITTARIUS_A_STAR_2025
Index Slot: §1.11 Black Hole Physics & Hawking Radiation, PAPER_086

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun□Sgr A* system at 27,000 ly, the validator test_Ug4_validation.py computes Ug4 = 3.352941×10^{22} J/m³ at t=0. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^4 day?¹, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole□stellar system boundary □ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{decay}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_bh	8.55×10^6 kg	EHT 2024-25

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
Orbital distance	d_g	2.55×10^{20} m	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_AGN	1.0 (quiescent)	Amplification factor
Decay constant	a	?/t_orb	Tied to ? = 0.0005/day
SCm density	[SCm]	0.99	Superconductive mode
UA density	[UA]	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun-Sgr A*** system at $t=0$, $t_n=0$, $A_{\text{AGN}}=1.0$:

$$Ug_4^{\text{Sun-SgrA}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: e^{-at}

The Ug_4 baseline decays exponentially with the UQFF ? parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-8} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-5}	1.44×10^{13}

Test case 2 (temporal decay e^{-at}) **PASS** (Ug_4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug_4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) □ PASS (U_{g4} increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: $\cos(\text{pt_n})$

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent U_{g4} oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum U_{g4})
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle $\cos(\text{pt_n})$) □ PASS

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_{bh} (kg)	d_{g} (m)	$U_{\text{g4}}(t=0)$ (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) □ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use U_{g4} :

- U_{g4} passes as a BuoyantForce component into the full $F_{\text{U_Bi_i}}$ master equation
 - Real-time U_{g4} evaluation: triggered by change in M_{bh} , d_{g} , or t_n
 - Output matches standalone `Ug4StarBlackHoleCalculator` to 10 significant figures
-

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	$U_{\text{g4}} = 3.352941 \times 10^{22}$ at ($t=0$, $t_n=0$)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	$A_{\text{AGN}} \times f_{\text{SCm}}$ amplification	PASS

Test Case	Physical Phenomenon	Result
4. Negative time	Ug4 > baseline (pre-collapse regime)	PASS
5. Temporal cycle	cos(pt_n) recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full F_U_Bi_i	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: `test_Ug4_validation.py` | `Ug4StarBlackHoleCalculator` | `SAGITTARIUS_A_STAR_2025`
| 7 tests PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun–Sgr A* system at 27,000 ly, the validator `test_Ug4_validation.py` computes $Ug4 = 3.352941 \times 10^{22} \text{ J/m}^3$ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^{\tau_1}$, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole–stellar system boundary – analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From `Ug4StarBlackHoleCalculator` and `UQFFConstantsDatabase`:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{\text{decay}}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_bh	$8.55 \times 10^36 \text{ kg}$	EHT 2024-25
Orbital distance	d_g	$2.55 \times 10^{20} \text{ m}$	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_AGN	1.0 (quiescent)	Amplification factor
Decay constant	a	?/t_orb	Tied to $\tau = 0.0005/\text{day}$

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
SCm density	[SCm]	0.99	Superconductive mode
UA density	[UA]	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun-Sgr A*** system at $t=0$, $t_n=0$, $A_{AGN}=1.0$:

$$Ug_4^{\text{Sun-SgrA}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: $e^{-\kappa t}$

The Ug4 baseline decays exponentially with the UQFF κ parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-8} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay $e^{-\kappa t}$) $\square$ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{AGN}(A_{AGN}) = A_{AGN} \cdot \left(1 + \frac{[SCm]}{10}\right)$$

For Sgr A* in quiescent state ($A_{AGN} = 1.0$, $[SCm] = 0.99$):

$$f_{AGN} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{AGN} = 3.5$ estimated):

$$f_{AGN} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) $\square$ **PASS** (Ug4 increases proportional to $A_{AGN} \times f_{SCm}$)

6. Temporal Cycle: $\cos(\text{pt_n})$

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle $\cos(\text{pt_n})$) $\square$ PASS

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_bh (kg)	d_g (m)	Ug4(t=0) (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) $\square$ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full $F_{U_{Bi_i}}$ master equation
 - Real-time Ug4 evaluation: triggered by change in M_{bh} , d_g , or t_n
 - Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures
-

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	Ug4 = 3.352941×10^{22} at (t=0, $t_n=0$)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	$A_{AGN} \times f_{SCM}$ amplification	PASS
4. Negative time	Ug4 > baseline (pre-collapse regime)	PASS
5. Temporal cycle	$\cos(\text{pt_n})$ recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full $F_{U_{Bi_i}}$	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: `test_Ug4_validation.py` | Ug4StarBlackHoleCalculator | SAGITTARIUS_A_STAR_2025
| 7 tests PASS .Groups[1].Value

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun–Sgr A* system at 27,000 ly, the validator `test_Ug4_validation.py` computes $Ug4 = 3.352941 \times 10^{22} \text{ J/m}^3$ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, *M87*, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}$ ¹, $[SCm] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole–stellar system boundary – analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From `Ug4StarBlackHoleCalculator` and `UQFFConstantsDatabase`:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{\text{decay}}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_{bh}	$8.55 \times 10^6 \text{ kg}$	EHT 2024-25
Orbital distance	d_g	$2.55 \times 10^{20} \text{ m}$	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_{AGN}	1.0 (quiescent)	Amplification factor
Decay constant	a	τ/t_{orb}	Tied to $\tau = 0.0005/\text{day}$
SCm density	$[SCm]$	0.99	Superconductive mode
UA density	$[UA]$	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun–Sgr A*** system at $t=0$, $t_n=0$, $A_{AGN}=1.0$:

$$Ug_4^{\text{Sun-SgrA}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: $e^{-\kappa t}$

The Ug4 baseline decays exponentially with the UQFF κ parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-7} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay $e^{-\kappa t}$) $\square$ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) $\square$ **PASS** (Ug4 increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: cos(pt_n)

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle cos(pt_n)) ☒ **PASS**

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_bh (kg)	d_g (m)	Ug4(t=0) (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) ☒ **PASS**

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full $F_{U_{Bi}i}$ master equation
 - Real-time Ug4 evaluation: triggered by change in M_{bh} , d_g , or t_n
 - Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures
-

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	Ug4 = 3.352941×10^{22} at (t=0, $t_n=0$)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	$A_{AGN} \times f_{SCM}$ amplification	PASS
4. Negative time	Ug4 > baseline (pre-collapse regime)	PASS
5. Temporal cycle	cos(pt_n) recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full $F_{U_{Bi}i}$	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: `test_Ug4_validation.py` | `Ug4StarBlackHoleCalculator` | `SAGITTARIUS_A_STAR_2025`
| 7 tests PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun–Sgr A* system at 27,000 ly, the validator test Ug4_validation.py computes $Ug_4 = 3.352941 \times 10^{22} \text{ J/m}^3$ at $t=0$. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e^{-at}), AGN feedback amplification, temporal cycle modulation ($\cos(pt_n)$), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87, Cygnus X-1).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^{\tau^1}$, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole–stellar system boundary – analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$Ug_4(M_{bh}, d_g, t, t_n, A_{AGN}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{bh}^2}{c^4 d_g^6} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{decay}(t) \cdot f_{AGN}(t_n, A_{AGN})$$

Parameter Definitions

Parameter	Symbol	Value (Sgr A*)	Physical Meaning
BH mass	M_bh	$8.55 \times 10^36 \text{ kg}$	EHT 2024-25
Orbital distance	d_g	$2.55 \times 10^{20} \text{ m}$	27,000 ly
Temporal UQFF	t_n	0.0 ? varied	UQFF normalized time
AGN amplitude	A_AGN	1.0 (quiescent)	Amplification factor
Decay constant	a	τ/t_{orb}	Tied to $\tau = 0.0005/\text{day}$
SCm density	[SCm]	0.99	Superconductive mode
UA density	[UA]	0.0001	Universal Antagonist
Cosmic time	t	0 ? 8	Physical time progression

3. Baseline Result

For the **Sun–Sgr A*** system at $t=0$, $t_n=0$, $A_{AGN}=1.0$:

$$Ug_4^{\text{Sun-SgrA}^*}(t = 0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS_A_STAR_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: $e^{-\kappa t}$

The Ug4 baseline decays exponentially with the UQFF κ parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-5} \text{ s}^{-1}$:

t (years)	f_decay	Ug4 (J/m ³)
0	1.000	3.353×10^{22}
1,000	0.833	2.793×10^{22}
10,000	0.163	5.472×10^{21}
100,000	4.3×10^{-10}	1.44×10^{13}

Test case 2 (temporal decay $e^{-\kappa t}$) $\square$ **PASS** (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10} \right)$$

For Sgr A* in quiescent state ($A_{\text{AGN}} = 1.0$, $[\text{SCm}] = 0.99$):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87* in jet-active state ($A_{\text{AGN}} = 3.5$ estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) $\square$ **PASS** (Ug4 increases proportional to $A_{\text{AGN}} \times f_{\text{SCm}}$)

6. Temporal Cycle: $\cos(\pi t_n)$

The UQFF normalized time $t_n = t/t_{\text{orbital}}$ creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At $t_n = 0$ (closest approach): $f_{\text{cycle}} = 1.0$ (maximum Ug4)
- At $t_n = 1$ (half-orbit): $f_{\text{cycle}} = 0.0$ (minimum)
- At $t_n = 2$ (full orbit): $f_{\text{cycle}} = 1.0$ (maximum again)

Test case 5 (temporal cycle $\cos(\text{pt}_n)$) $\square$ PASS

7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

System	M_{bh} (kg)	d_g (m)	$Ug4(t=0)$ (J/m ³)
SGR_A_STAR_SYSTEM	8.55×10^{36}	2.55×10^{20}	3.353×10^{22}
M87_STAR_SYSTEM	$\sim 1.2 \times 10^{40}$	$\sim 5 \times 10^{23}$	$\sim 6.8 \times 10^{17}$
CYGNUS_X1_SYSTEM	$\sim 1.4 \times 10^{31}$	$\sim 5.7 \times 10^{17}$	$\sim 1.2 \times 10^{25}$

Test case 7 (all 3 predefined systems) $\square$ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full $F_{U_{Bi_i}}$ master equation
- Real-time Ug4 evaluation: triggered by change in M_{bh} , d_g , or t_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

Test Case	Physical Phenomenon	Result
1. Baseline	$Ug4 = 3.352941 \times 10^{22}$ at ($t=0$, $t_n=0$)	PASS
2. Temporal decay	$e^{(-at)}$? monotonic decrease	PASS
3. AGN feedback	$A_{\text{AGN}} \times f_{\text{SCm}}$ amplification	PASS
4. Negative time	$Ug4 >$ baseline (pre-collapse regime)	PASS
5. Temporal cycle	$\cos(\text{pt}_n)$ recurrence ? [0,1]	PASS
6. CP2 integration	Consistent with full $F_{U_{Bi_i}}$	PASS
7. All 3 systems	SGR_A, M87, CYG_X1 all finite	PASS

Source: `test_Ug4_validation.py` | `Ug4StarBlackHoleCalculator` | `SAGITTARIUS_A_STAR_2025`
 | 7 tests PASS

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \times \exp(-? \times ?t) = 1 - 5.7e-1 \times \exp(-2.9e-4) = 4.3e-1$; F_U at event horizon = $2.0e+18$ m/s².

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \hat{\mu}\text{ s}^{-1} \text{ day}^{-1} \hat{\mu}\text{ s}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	$0.60 \hat{\mu}\text{ s}^{-1} \text{ day}^{-1}$	Buoyancy coupling coefficient
$\hat{k}_{DPM}$	1.5	Ug1 DPM-dipole coupling
$\hat{k}_{out}$	1.2	Ug2 outer-bubble charge coupling
$\hat{k}_{str}$	1.8	Ug3 string-rotation coupling
$\hat{k}_{vac}$	2.0	Ug4 vacuum-concentration coupling
$\hat{I}$	$10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1}$	Inertia tensor scale
$E_{\text{react}}(0)$	$10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1} \text{ J}$	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{\mu}\text{ s}^{-1}$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} \text{igr}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1}$	4th Dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{\Gamma}^0 = 10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1}$, $\hat{\Gamma}^2_i = 10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1}$, $\hat{k}_{DPM} = 10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1}$, $\hat{k}_{out} = 10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c) \text{igr}) \text{imesigl}(1 + A_q \cos(\Delta\omega t) \text{igr})$$

Symbol	Value	Description
$\hat{\Gamma}^0_c$	$10 \hat{\mu}\text{ s}^{-1} \hat{\mu}\text{ s}^{-1} \text{ kg/m}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{\Gamma}^0$	$2 \hat{\mu}\text{ s}^{-1} / (434 \hat{\mu}\text{ s}^{-1} \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{a}_i$)	Multi-scale field interactions
Buoyant	$\hat{I}_i \tilde{\Delta} Ubi$	Expanding nebulae, stellar winds
Superconductive	$\tilde{\Delta} (1 + 10 \hat{A}^1 \hat{A}^3 \hat{A} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.